

# ARASH HEIDARI

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## EDUCATION

**Ghent University** **Ghent, Belgium**  
**Computer Science (Ph.D)** **Aug 2021 – Aug 2025**

- **Supervisors:** Dr. Ivo Couckuyt, Dr. Sebastian Rojas Gonzalez
- **Thesis:** Targeting Evaluation, Computation and Interaction Costs in Expensive Black-Box Multi-Objective Optimization

**Iran University of Science and Technology** **Tehran, Iran**  
**Electrical Engineering (M.Sc.)** **Sep 2016 – Jul 2018**

- **Supervisor:** Dr. Sattar Mirzakuchaki
- **Thesis:** DPA-Resistant and Low Power Cryptography in IoT Devices.
- **GPA:** 18.27 / 20

**Iran University of Science and Technology** **Tehran, Iran**  
**Electrical Engineering (B.Sc.)** **Sep 2012 – Jul 2016**

- **Supervisor:** Dr. Sattar Mirzakuchaki
- **Thesis:** Internet-Enabled Control for Enhanced Surveillance System Operation
- **GPA:** 16.19 / 20

## EXPERIENCE

**Ruhr University Bochum** **Bochum, Germany (Remote from Belgium)**  
**Postdoctoral Researcher** **Apr 2026 – Present**

- Spearheading advanced research projects on multi-agent algorithmic recourse, designing robust methodologies for complex machine learning systems.
- Developing novel algorithmic recourse frameworks targeted at top-tier AI and machine learning venues, including AAAI, AISTATS, and NeurIPS.

**IDLab (imec - Ghent University)** **Ghent, Belgium**  
**AI Researcher** **Aug 2021 – Jul 2025**

- Execution of various AI-based projects from diverse corporate clients at IDLab.
- Successful collaborations with industry leaders: Boeing, Sirris, and Atlas Copco.
- Involvement in University-Industry Collaboration (Flanders AI Research Program).
- Supervision of junior researchers and master students.

**Iran University of Science and Technology** **Tehran, Iran**  
**AI Researcher** **Feb 2019 – Dec 2020**

- Execution of AI-based projects, as the primary responsibility.
- Collaborations with PhD and master's students on AI-related projects.
- Teaching machine learning, deep learning, and Python programming language.

## SKILLS

**Machine Learning** Deep Learning, Time-Series Analysis, Optimization, Computer Vision, Uncertainty Quantification, Anomaly Detection, Resource Efficient ML, Physics-Informed ML, GANs, Bayesian ML, GNNs, Algorithmic Recourse, Explainable AI

**Programming** Python, C/C++, JavaScript, R, TensorFlow, PyTorch, SQL, FastAPI, Plotly

**Software Development** GUI Development, Embedded Software Development, Linux

**MLOps** MLflow, Git, Docker, Grafana

**Languages** Persian (C2), English (C1), German (B1), Dutch, Spanish, French (A2)

**Soft Skills** Leadership, Interdisciplinary Collaboration, Public Speaking, Technical Communication, Adaptability, Curiosity-Driven Learning

## SELECTED RESEARCH & ENGINEERING PROJECTS

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### Trustworthy AI for Aerospace Certification 2022-2025

- Led a multi-year R&D project in collaboration with Boeing to build trustworthy surrogate models for a safety-critical system, targeting FAA certification.
- Developed and validated high-stakes time-series forecasting model architectures using CNN, LSTM, and Transformers.
- Designed and implemented the core "trustworthy" framework, using Deep Ensembles and Quantile Regression for robust Uncertainty Quantification (UQ) and advanced Anomaly Detection pipelines.
- Built a GUI dashboard that translated complex UQ/failure analyses into actionable insights for engineers, which was key to the project's multi-year extension.

### Predictive Maintenance for Renewable Energy 2021-2023

- Developed a high-fidelity data-driven surrogate model of a wind turbine's physical behavior in collaboration with Sirris from complex sensor data to enable robust, high-speed simulation.
- Designed a multi-stage predictive maintenance system using anomaly detection methods (e.g., AutoEncoders, One-Class SVMs) to predict equipment failures.
- Implemented representation learning techniques, including Triplet Loss, to map complex sensor states to an informative feature space, improving anomaly separability.

### Data-Efficient Preference-Driven Multi-Objective Optimization 2021-2025

- Published novel research (GECCO, IEEE TEVC) on algorithms for "expensive" black-box optimization problems where each evaluation is costly.
- Specialized in Bayesian Optimization and Surrogate-Assisted Evolutionary Algorithms (SA-EAs) to find data-efficient solutions.
- Pioneered preference-driven, interactive methods that learn a user's goals in real-time, allowing the algorithm to converge directly to the most preferred region of the Pareto front instead of finding the entire, unmanageable front.

### Biomedical Computer Vision: Retinal Pathology Classification 2019-2020

- Developed Computer Vision models for the automated classification of retinal pathologies, such as Diabetic Macular Edema, from Optical Coherence Tomography scans.
- Researched and implemented Generative Adversarial Networks (GANs) to augment the scarce, imbalanced medical imaging datasets, improving model robustness and performance.

## PUBLICATIONS

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### Authored 10+ publications in leading scientific venues 2019-2026

- Authored peer-reviewed contributions in premier artificial intelligence and engineering venues, including AAAI and IEEE Transactions.
- Serving as a Program Committee member and reviewer for leading international conferences, including AAAI.
- Full publication list available upon request.